Course Outline

MAT 112 : Linear Algebra and Differential Equations

1. **Algebra of Matrices:**

Matrices, Transpose of a matrix, Square matrices, Powers of matrices, Polynomials in matrices, Invertible (Nonsingular) matrices, special types of square matrices, Complex matrices, and Block matrices.

2. **Basic Operations:**

Matrix addition and Scalar multiplication, Matrix multiplication, Calculation of Inverse, Row-echelon form, Elementary row and column operations, Rank.

Simultaneous Linear Equations: Augmented matrix, Gaussian elimination method, Pivoting strategies, Gauss-Jordan elimination, and LU decomposition.

Class Test 01: On Topic 1 and 2.

3. Determinants:

Determinant of a square matrix, Minor, Cofactor, Inverse of a matrix using determinant, Pivotal condensation.

4. Vectors:

Dimension, Linear dependence and independence, Linear combinations.

Eigenvalues and Eigenvectors: Characteristic equations, Properties of eigenvalues and eigenvectors, Linearly independent eigenvectors, Cayley-Hamilton theorem.

5. Complex Numbers:

The complex number system, Fundamental operations with complexnumbers, Absolute value, axiomatic foundation of the complex number system, Graphical representation of complex numbers, Polar form of complex numbers, De Moivre’s theorem, Roots of complex numbers, Euler’s formula, Polynomial equations, The nth roots of unity, Vector interpretation of complex numbers, Dot and cross product, Complex conjugate coordinates.

Class Test 02: On Topic 3, 4 and 5

Mid-Term Syllabus: 1,2,3,4,5.

6. Differential Equations:

6.1 Ordinary differential equation: formation of differential equations; solution of first order differential equations by various methods; solution of differential equation of first order but higher degrees; solution of general linear equations of second and higher orders with constant co-efficient; solution of Euler's homogeneous linear differential equations.

6.2 Partial differential equation: introduction, linear and non-linear first order differential equations; standard forms; linear equations of higher order; equations of the second order with variable coefficients.

7. Fourier Series and its Applications:

The need for Fourier series, Periodic functions, Piecewise continuous functions, Definition of Fourier series, Dirichlet's conditions, Odd and even functions, Half-range Fourier sine or cosine series, Parseval's identity, Uniform convergence, Integration and differentiation of Fourier series, Complex notation for Fourier series, Double Fourier series, Applications of Fourier series.

Syllabus for Final: 6,7 (One or two topics from mid may be included)